

Stability to post-hoc UQ without recalibrating $\mathcal{C}$

SURA research note (working)

27 July 2026

Scope ceiling (Wenbin): UQ (1)(2); selected object $\hat{\theta}(\mathcal{D})$ only; goal (iii) = data-randomize selection (**not** recalibrate $\mathcal{C}$).

Setup. Fix classical level α and sets $\mathcal{C}^{(\alpha)}(X; \mathcal{D}, \theta)$ (writeup (1) arity: third slot = θ ; α only as level / superscript — not in the third slot). Let $\hat{\theta}(\mathcal{D})$ come from constrained opt (2) after a randomized selection design (RNM / soft-argmin). If $\hat{\theta}$ is $(\eta, 0, \nu)$ -stable (ZJ), then

$$\text{Infl} := P(Y \notin \mathcal{C}^{(\alpha)}(X; \mathcal{D}, \hat{\theta})) - \alpha \leq (e^\eta - 1)\alpha + \nu.$$

Sending $\eta, \nu \rightarrow 0$ restores (1) at the *same* α and the *same* map $\mathcal{C}$.

Part I (design). Concentrate $(\hat{f}, \hat{g})$ at rate $\varepsilon(n, \nu)$ (DKW). Choose Lap scale $b = 2\varepsilon/\eta$ or soft $\beta = \eta/(2\varepsilon) \Rightarrow (\eta, 0, \nu)$ -stable $\tilde{\theta}$, **restricted to** Θ_{val} (score-threshold: $\theta \geq F^{-1}(1 - \alpha)$; tradeoff: conservative Θ_0^ε). $\eta \rightarrow 0$ costs selection utility (not free).

Instances of (2). ($\mathbf{2}_{\text{nest}}$): $\hat{f} = 1 - \hat{F}_n$, box $\hat{g}$ — hard argmin collapses to $\theta = 1$; live object is randomized. ($\mathbf{2}_{\text{trade}}$): $\hat{f}(\theta) = \theta$, $\hat{g} = \text{misc} - \alpha_0$ — hard $\hat{\theta} = \text{empirical quantile}$ (non-degenerate). Constraint α_0 is not the classical level α of (1); numerics set $\alpha_0 = \alpha$.

Part II. Set-valued ZJ transfer under (OV)+(ST) (writeup (1) on $\text{supp}(A_0)$; for score-threshold sets, only $\theta \geq F^{-1}(1 - \alpha)$): keep classical level α fixed; bound Infl as above. Recalibrating to level $\alpha e^{-\eta}$ answers a different question (rejected by W5). Notation: $\mathcal{C}^{(\alpha)}$ = writeup $\mathcal{C}$ at level α ; third slot remains θ .

Numerics ($(\mathbf{2}_{\text{trade}})$, $\eta = \varepsilon$, $\nu = 0.05$, $\alpha_0 = \alpha = 0.1$).

n	ε	Infl bound	misc MC	$\widehat{\text{Infl}}$
100	0.136	0.065	0.123	+0.023
400	0.068	0.057	0.086	−0.015
1600	0.034	0.053	0.064	−0.036
6400	0.017	0.052	0.056	−0.044

Bound decreases in n ; MC miscoverage stays within α +bound. Artifacts: **research/formal/***, demos under **research/experiments/**, audits ALL PASS.

Ask. Does this chain (stability design $\rightarrow$ Infl at fixed $\mathcal{C}^{(\alpha)}$ on $(\mathbf{2}_{\text{trade}})$) match the intended SURA target, or should (iii) emphasize a different randomization?